

GABA and Glutamate

Glutamine-glutamate cycle (carriers SLC1A2/SLC1A3, GLS, GLUL, AMPA, NMDA and metabotropic receptors, CACNA1C) and GABA (GAD1/GAD2, GABA-A, ABAT, ALDH5A1, SLC6A1)

Sérgio F · 03-10-2026 · local time

Informative and educational genetic report, based only on Sérgio F genes (genotyping chip + local imputation). It is neither a diagnosis nor a prescription. All actions are marked as suggestion (not applied).

Contents

Section	Page
1. Summary	3
2. How to read this report	4
3. SNP genotypes at SNP	5
4. Modified genes and typical genes	8
5. Reading as a system	9
6. Mechanisms of ageing	10
7. Practical suggestions	11
8. Limitations	12
9. Methods and sources	13
10. Annex: all variants analysed	14

1. Summary

What was seen. 34 SNPs were read in 22 genes (19 of the glutamine-glutamate cycle and 15 of the GABA cycle): 18 of the chip, 14 imputed with $DR2 \geq 0.8$ and 2 not found (**GRM7** rs3749380 and **SLC6A1** rs2697153). As a rule of Sérgio F, the results of neurotransmitters are descriptive, without judgment of good or bad.

Glutamate. The removal of glutamate by the astrocytes has an Increased label in **SLC1A3** (C/C) and mixed in **SLC1A2** (2 SNPs, both 1 Increased + 1 Reduced). Recycling has the **GLUL** Reduced (C/C) and the **GLS** Altered in the 2 SNPs (imputed). In the receivers, **GRIN2B** (NMDA) is Increased in the 2 SNPs, **GRIA1** (AMPA) is Increased, **GRIN2A** is mixed and the **GRM3** metabotropic is Reduced in the 3 SNPs (concordants). The **CACNA1C** rs1006737 has Altered 1 (A) and the **SLC6A15** is Reduced. **GRIA3** (X) is Typical.

GABA. The synthesis has the **GAD1 Reduced** in the 2 SNPs and the GAD2 mixed in the 2. The degradation has the **ABAT** Reduced (imputed) and the **ALDH5A1** Typical. In GABA-A, **GABRA1**, **GABRA2** and **GABRB3** rs4906902 receptors are Increased; the GABRB3 rs2081648 is Reduced; the **GABRA6** is mixed; **GABRB2** and **GABRG2** are Typical.

What does it mean? They are small and added, non-diagnoses. The descriptive portrait is of excitatory NMDA/AMPA signaling with Increased tags and Reduced metabotropic retroaction (GRM3), with astrocytic removal of preserved or increased glutamate, and a synthesis of GABA (GAD1) with Reduced label compensated by slower degradation (ABAT) and GABA-A receptors with several Increased tags. What most modulates these pathways is known: sleep, exercise, adequate stress, magnesium and zinc management and zero alcohol.

2. How to read this report

- **Data:** Sérgio F chip genotypes (GRCh38, strand +) and, when SNP is not in the chip, locally imputed genotypes (Beagle 5.5, public panel of genomes complete to 30x) with quality $DR2 \geq 0.8$. Under this threshold, or outside the imputed zone, SNP is treated as SNP not found in the data.
- **Variants:** 34 SNPs: 18 of the chip, 14 imputed and 2 not found. All genotypes were reread directly in the raw DNA and in the imputation for this report.
- **Strand:** All genotypes are in the GRCh38 strand +. 0 SNPs needed conversion to complementary strand and 0 are palindromic (A/T or C/G). When the literature uses the strand of the gene, the letters may differ from those of this report.
- **Chromosome X:** Sérgio F has only one copy of X outside the pseudoautosomal regions, so the SNPs of X appear with a single letter (hemizygous).
- **No good/bad judgment:** by Sérgio F rule for neurotransmitters, variants are not classified as favourable or unfavourable. The activity tags of the original report are used (in English: Increased, Reduced, Typical, Altered, Normal, Higher, Lower). The pathways are read as altered (different from Typical/Normal) or typical.
- **Heterozygous:** show the two alleles and the effect of each one; no new label is invented.

3. SNP genotypes at SNP

For each SNP: genotype (strand +; GRIA3 with one letter), the effect of each allele according to the origin report, the global activity, the health theme and the Enhancers (potentiators) indicated. The origin report does not indicate Blockers for any of these genes, so it appears "—".

Glutamine-glutamate cycle

Biological order: glutamate reuptake by astrocytes (SLC1A2/EAAT2, SLC1A3/EAAT1), glutamine ↔ glutamate recycling (GLUL glutamine form and remove ammonia; GLS is back to form glutamate, AMPA (GRIA1, GRIA3), NMDA (GRIN2A, GRIN2B) and metabotropic receptors (GRM3, GRM7), CACNA1C calcium channel and SLC6A15 transporter. GRIA3 is in the X chromosome (one letter).

Genotype and activity

Gene	SNP	Genotype	1 allele → effect	2 allele → effect	Global activity
GRIA1	rs2195450	A/A	A → Increased activity	A → Increased activity	Increased activity (2 copies)
SLC1A2	rs4354668	T/G	T → Increased activity	G → Reduced activity	Mixed: 1 Enhanced Activity + 1 Reduced Activity
SLC1A2	rs3794087	G/T	G → Increased activity	T → Reduced activity	Mixed: 1 Enhanced Activity + 1 Reduced Activity
SLC1A3	rs2269272	C/C	C → Increased activity	C → Increased activity	Increased activity (2 copies)
GLS	rs12185688	T/T	T → Altered activity	T → Altered activity	Altered activity (2 copies)
GLS	rs13000464	C/C	C → Altered activity	C → Altered activity	Altered activity (2 copies)
GRIA3	rs687577	C (hemizygous, X)	C → Typical activity	— (1 copy: X chromosome)	Typical activity (1 copy, X)
GRIN2A	rs11644461	C/T	C → Altered activity	T → Normal activity	Mixed: 1 Altered Activity + 1 Normal Activity
GRIN2A	rs7206256	A/G	A → Altered activity	G → Normal activity	Mixed: 1 Altered Activity + 1 Normal Activity
GRIN2B	rs1019385	A/A	A → Increased activity	A → Increased activity	Increased activity (2 copies)
GRIN2B	rs1805476	T/T	T → Increased activity	T → Increased activity	Increased activity (2 copies)
GRM3	rs12704290	G/G	G → Reduced activity	G → Reduced activity	Reduced activity (2 copies)
GRM3	rs6465084	A/A	A → Reduced activity	A → Reduced activity	Reduced activity (2 copies)
GRM3	rs274622	T/T	T → Reduced activity	T → Reduced activity	Reduced activity (2 copies)
GRM7	rs3749380	SNP not found in the data	SNP not found in the data	SNP not found in the data	SNP not found in the data
GRM7	rs779867	T/C	T → Increased activity	C → Reduced activity	Mixed: 1 Enhanced Activity + 1 Reduced Activity
CACNA1C	rs1006737	G/A	G → Normal activity	A → Altered activity	Mixed: 1 Normal Activity + 1 Altered Activity
GLUL	rs10911021	C/C	C → Reduced activity	C → Reduced activity	Reduced activity (2 copies)
SLC6A15	rs1545843	A/A	A → Reduced activity	A → Reduced activity	Reduced activity (2 copies)

Health theme, Enhancers and Blockers

Gene	SNP	Health theme	Enhancers	Blockers
GRIA1	rs2195450	Excessive glutamate activity and migraine risk	—	—
SLC1A2	rs4354668	mood and nervous disorders	vitamin C, NAC	—
SLC1A2	rs3794087	mood and nervous disorders	vitamin C, NAC	—
SLC1A3	rs2269272	protection against excess glutamate, stress, mood and attention	vitamin C	—
GLS	rs12185688	Glutamine/glutamate ratio and mental health	—	—

Gene	SNP	Health theme	Enhancers	Blockers
GLS	rs13000464	Glutamine/glutamate ratio and mental health	—	—
GRIA3	rs687577	glutamate activity and mood	—	—
GRIN2A	rs11644461	glutamate activity and psychotic disorders	—	—
GRIN2A	rs7206256	glutamate activity and psychotic disorders	—	—
GRIN2B	rs1019385	cognition and obsessive-compulsive disorder	—	—
GRIN2B	rs1805476	cognition and obsessive-compulsive disorder	—	—
GRM3	rs12704290	additions and cognitive problems	—	—
GRM3	rs6465084	additions and cognitive problems	—	—
GRM3	rs274622	additions and cognitive problems	—	—
GRM7	rs3749380	Mood and cognitive disorders	NAC	—
GRM7	rs779867	Mood and cognitive disorders	NAC	—
CACNA1C	rs1006737	GABA/glutamate balance, psychotic disorders and anxiety	—	—
GLUL	rs10911021	elimination of ammonia and glutamate, and metabolic health	magnesium, zinc	—
SLC6A15	rs1545843	BCAA and neurotransmitter transport, and mood	—	—

GABA cycle

Biological order: synthesis of GABA from glutamate (GAD1, GAD2; depends on vitamin B6), GABA-A receptors (GABRA1, GABRA2, GABRA6, GABRB2, GABRB3, GABRG2), reuptake (SLC6A1/GAT1) and degradation (ABAT, then ALDH5A1).

Genotype and activity

Gene	SNP	Genotype	1 allele → effect	2 allele → effect	Global activity
GABRA2	rs279858	T/T	T → Increased activity	T → Increased activity	Increased activity (2 copies)
GABRA6	rs3219151	C/T	C → Increased activity	T → Reduced activity	Mixed: 1 Enhanced Activity + 1 Reduced Activity
GABRG2	rs211037	C/C	C → Typical activity	C → Typical activity	Typical activity (2 copies)
GAD1	rs1978340	A/A	A → Reduced activity	A → Reduced activity	Reduced activity (2 copies)
GAD1	rs3749034	G/G	G → Reduced activity	G → Reduced activity	Reduced activity (2 copies)
ABAT	rs2270288	T/T	T → Reduced activity	T → Reduced activity	Reduced activity (2 copies)
ALDH5A1	rs2760118	C/C	C → Typical activity	C → Typical activity	Typical activity (2 copies)
GAD2	rs992990	C/A	C → Reduced activity	A → Increased activity	Mixed: 1 Reduced Activity + 1 Increased Activity
GAD2	rs2236418	A/G	A → Reduced activity	G → Increased activity	Mixed: 1 Reduced Activity + 1 Increased Activity
GABRB2	rs1816071	T/T	T → Typical activity	T → Typical activity	Typical activity (2 copies)
GABRB2	rs1816072	A/A	A → Typical activity	A → Typical activity	Typical activity (2 copies)
GABRB3	rs2081648	T/T	T → Reduced activity	T → Reduced activity	Reduced activity (2 copies)
GABRB3	rs4906902	A/A	A → Increased activity	A → Increased activity	Increased activity (2 copies)
SLC6A1	rs2697153	SNP not found in the data	SNP not found in the data	SNP not found in the data	SNP not found in the data
GABRA1	rs2279020	A/A	A → Increased activity	A → Increased activity	Increased activity (2 copies)

Health theme, Enhancers and Blockers

Gene	SNP	Health theme	Enhancers	Blockers
GABRA2	rs279858	Additive behavior (risk of alcohol use)	apigenin	—
GABRA6	rs3219151	anxiety and stress response, and alcohol dependence	—	—
GABRG2	rs211037	anxiety and seizures	zinc, magnesium	—
GAD1	rs1978340	GABA production, attention problems and additions	zinc	—

Gene	SNP	Health theme	Enhancers	Blockers
GAD1	rs3749034	GABA production, attention problems and additions	zinc	—
ABAT	rs2270288	GABA metabolism and cognitive health	vitamin B6	—
ALDH5A1	rs2760118	elimination of toxic metabolites and cognitive health	—	—
GAD2	rs992990	Mental health and appetite control	—	—
GAD2	rs2236418	Mental health and appetite control	—	—
GABRB2	rs1816071	Psychotic disorders	—	—
GABRB2	rs1816072	Psychotic disorders	—	—
GABRB3	rs2081648	Cognitive and mood disorders	—	—
GABRB3	rs4906902	Cognitive and mood disorders	—	—
SLC6A1	rs2697153	GABA reuptake and anxiety risk	—	—
GABRA1	rs2279020	mood problems and seizures	magnesium, apigenin	—

4. Modified genes and typical genes

Genes of glutamate with different result from Typical/Normal

- **SLC1A3 (EAAT1)**: Increased (rs2269272 C/C).
- **SLC1A2 (EAAT2)**: mixed in 2 SNPs (rs4354668 T/G imputed; rs3794087 G/T): 1 Increased + 1 Reduced.
- **GLS**: Altered in 2 SNPs (rs12185688 T/T, rs13000464 C/C; imputed).
- **GLUL**: Reduced (rs10911021 C/C).
- **GRIA1 (AMPA)**: Increased (rs2195450 A/A; imputed).
- **GRIN2A (NMDA)**: mixed in 2 SNPs (rs11644461 C/T, rs7206256 A/G; imputed): 1 Altered + Normal 1.
- **GRIN2B (NMDA)**: Increased in 2 SNPs (rs1019385 A/A; rs1805476 T/T imputed).
- **GRM3**: Reduced in 3 SNPs (Imputed rs12704290 G/G, rs6465084 A/A, rs274622 T/T).
- **GRM7**: rs779867 mixed (T/C: 1 Increased + 1 Reduced; imputed).
- **CACNA1C**: 1 Normal + 1 Altered (rs1006737 G/A).
- **SLC6A15**: Reduced (rs1545843 A/A).

Genes of GABA with different result from Typical/Normal

- **GAD1**: Reduced in 2 SNPs (rs1978340 A/A, rs3749034 G/G).
- **GAD2**: mixed in 2 SNPs (rs992990 C/A, rs2236418 A/G): 1 Reduced + 1 Increased.
- **ABAT**: Reduced (rs2270288 T/T; imputed).
- **GABRA1**: Increased (rs2279020 A/A).
- **GABRA2**: Increased (rs279858 T/T).
- **GABRA6**: Mixed (rs3219151 C/T: 1 Increased + 1 Reduced).
- **GABRB3**: rs2081648 Reduced (T/T) and rs4906902 Increased (A/A; both imputed).

Genes with Typical/Normal Result

- **Glutamate**: GRIA3 (rs687577 C; X).
- **GABA**: GABRG2 (rs211037 C/C), GABRB2 (rs1816071 T/T, rs1816072 A/A; imputed), ALDH5A1 (rs2760118 C/C).

Unread

- **GRM7** rs3749380 and **SLC6A1** rs2697153: SNP not found in the data (outside the chip and without imputation with $DR2 \geq 0.8$).

Contradictory or non-concordant SNPs

- **GABRB3**: The 2 SNPs have opposite tags (Red and Increased). They are different sites of the gene; the balance is indeterminate.
- **GRM7**: only 1 of the 2 SNPs was read.

5. Reading as a system

Read as a whole, without judgment: 1. Removal of glutamate from synapse: SLC1A3 has Increased tag and SLC1A2 is mixed. It describes a preserved or increased astrocytic cleaning of glutamate. 2. Glutamine-glutamate recycling: GLUL Reduced (glutamine formation and ammonia removal) and GLS Altered (glutamate formation from glutamine). With the GLS already read (before missing), the cycle is complete: the two senses have different Typical tags. 3. Excitatory receivers: NMDA GRIN2B Increased (2 SNPs concordant), AMPA GRIA1 Increased, GRIN2A mixed. The presynaptic metabotropic brake GRM3 is re-educated in the 3 SNPs, the most consistent finding on the glutamate side. Together: Excitatory tone with tags in favor of more signaling, without clinical significance by itself. 4. Neuronal calcium: CACNA1C with Altered 1 allele (the rs1006737 allele, studied in mood and anxiety). 5. GABA synthesis: GAD1 Reduced (2 SNPs) and GAD2 mixed. GAD depends on vitamin B6 (P5P), already present in the stack. 6. GABA degradation: ABAT Reduced (slowest degradation) and ALDH5A1 Typical. A slower degradation can compensate part of a smaller synthesis. 7. GABA-A receivers: GABRA1, GABRA2 and GABRB3 (1 SNP) Increased; GABRA6 mixed; GABRB2 and GABRG2 Typical. GABRA2 rs279858 is the most studied SNP in alcohol response, which is already zero.

Connections to other reports: Serotonin and Melatonin, Dopamine and Noradrenaline, Anxiety and Stress Resilience, Mood and Depression Risk, Circadian Rhythm and Sleep, Migraine (GRIA1), Epilepsy and Excitability (when it exists).

6. Mechanisms of ageing

Mapping in 84 mechanisms of aging of Sérgio F immortality objectives (only the mechanisms affected by this report).

Mechanism	Genetic finding	Meaning
Mitochondrial Calcium Deregulation	Excitotoxicity: glutamate via NMDA (GRIN2B and GRIA1 Increased); brake GRM3 Reduced in 3 SNPs leads calcium to mitochondria; astrocytic removal SLC1A3 Increased, SLC1A2 mixed	without judgment (descriptive)
Intercellular Communication	GABA-A receivers with several Increased tags; GAD1 Reduced; ABAT Reduced	without judgment (descriptive)
Neuroinflammation	Extracellular glutamate and astrocytes (SLC1A2/SLC1A3, GLUL Reduced) bind excitability and inflammation of CNS	without judgment (descriptive)
Mitochondrial dysfunction	Glutamine-glutamate cycle and GABA shunt (ABAT, ALDH5A1) feed the neuronal Krebs cycle	without judgment (descriptive)
Sleep disturbance	GABA-A receptors (GABRA1, GABRA2, GABRB3) are the target of deep sleep and hypnotics	without judgment (descriptive)
Proteostasis loss	GLUL removes ammonia (glutamine form); Reduced label	without judgment (descriptive)
Oxidative stress	Excess of glutamate and calcium (CACNA1C 1 allele Altered) increase neuronal reactive species	without judgment (descriptive)

7. Practical suggestions

All suggestions are suggestion (not applied): the real stack of Sérgio F remains as it is, and any change should be discussed with the doctor.

1. **Regular and sufficient sleep:** deep sleep is the GABAergic state par excellence and reduces glutamate accumulated during the day. *Suggestion (not applied).*
2. **Regular exercise,** aerobic and strength: increases the astrocytic reuptake of glutamate and GABAergic tone. *Suggestion (not applied).*
3. **Magnesium and zinc** (Enhancers of GLUL, GABRG2, GABRA1 and GAD1), preferably for feeding (legumes, dried fruits, seeds, green vegetables); magnesium is also a physiological blocker of the NMDA receptor (GRIN2B Increased). *Suggestion (not applied).*
4. **Vitamin C** by feeding (Enhancer of SLC1A2 and SLC1A3) and **NAC**, already in the stack (Enhancer of SLC1A2 and GRM7). *Suggestion (not applied).*
5. **Vitamin B6** (GAD and ABAT cofactor): it is already in the stack; it does not increase without the doctor. *Suggestion (not applied).*
6. **Apigenin** appears as Enhancer of GABRA2/GABRA1; it is among the concentrated polyphenols whose pause was suggested because of the liver, so the preferred source is food (camomyl, parsley). *Suggestion (not applied).*
7. **Zero alcohol** (It is already the rule): GABRA2 rs279858 and GABRA6 are the most related genes to alcohol response. *Suggestion (not applied).*
8. **Stress management** (slow breathing, meditation, contact with nature): chronic stress increases glutamate and reduces GABA. *Suggestion (not applied).*
9. **GABAergic drugs (benzodiazepines, Z-drugs, gabapentinoids):** These SNPs do not predict the response; use only with the doctor and avoid combining alcohol or opioids. *Suggestion (not applied).*
10. **Supplements of GABA, pregnenolone and PEA** are not recommended without the doctor (weak evidence). *Suggestion (not applied).*

8. Limitations

- The effects of each SNP are small and the tags (Increased, Reduced, Altered...) come from the original report. They do not measure levels of GABA or glutamate in the brain.
- Two SNPs were not read (GRM7 rs3749380, SLC6A1 rs2697153); therefore the reuptake of GABA (SLC6A1) is missing.
- Fourteen SNPs come from imputation (DR2 0.85–1.00); SLC1A2 rs4354668 has the lowest DR2 (0.85).
- By Sérgio F rule, there is no favourable/unfavourable judgment in neurotransmitters.
- Conclusions only from genes; it is not a diagnosis.

9. Methods and sources

- **Genes and SNPs:** exactly the 22 genes and 34 SNPs of the original order (19 of glutamate, 15 of GABA), with the allele tags of the original report. Nothing was added or replaced; Enhancers were transcribed as indicated.
- **Genotypes:** Sérgio F chip (GRCh38, strand +), reread for this report. Absent SNPs come from local imputation (Beagle 5.5, public panel of complete 30x genomes), used only with $DR2 \geq 0.8$. No genetic data came out of the analysis computer; only public positions were consulted by rsID.
- **Guidance:** The alleles of the origin report are in the strand + (confirmed by VCF alleles and by the frequency in the reference population); there are no palindromic SNPs in this set.
- **Chromosome X:** GRIA3 out of pseudoautosomal regions → a letter (hemizygous).
- **Comparison with previous analysis (02-10-2026):** The 18 genotypes of the chip are equal; the 14 imputed confirm the values already integrated in the profile.
- **Context Literature:** GWAS Catalog, dbSNP and reviews on the glutamine-glutamate-GABA cycle (e.g. CACNA1C rs1006737: Ferreira 2008; GABRA2 and alcohol: Edenberg 2004).

10. Annex: all variants analysed

Via	Gene	SNP	Alleles of the report	Genotype	Source
Glutamine-glutamate cycle	GRIA1	rs2195450	A=Increased activity/ G=Reduced activity	A/A	imputation DR2=0.94
Glutamine-glutamate cycle	SLC1A2	rs4354668	G=Reduced activity/T=	T/G	imputation DR2=0.85
Glutamine-glutamate cycle	SLC1A2	rs3794087	G=Increased activity/ T=Reduced activity	G/T	chip
Glutamine-glutamate cycle	SLC1A3	rs2269272	C=Increased activity/ T=Reduced activity	C/C	chip
Glutamine-glutamate cycle	GLS	rs12185688	C=Typical activity/T=Altered activity	T/T	imputation DR2=0.99
Glutamine-glutamate cycle	GLS	rs13000464	C=Altered activity/T=Typical activity	C/C	imputation DR2=0.99
Glutamine-glutamate cycle	GRIA3	rs687577	A=Reduced activity/ C=Typical activity	C (hemizygous, X)	chip
Glutamine-glutamate cycle	GRIN2A	rs11644461	C=Altered activity/T=Normal activity	C/T	imputation DR2=0.99
Glutamine-glutamate cycle	GRIN2A	rs7206256	A=Altered activity/G=Normal activity	A/G	imputation DR2=0.99
Glutamine-glutamate cycle	GRIN2B	rs1019385	A=Increased activity/ C=Reduced activity	A/A	chip
Glutamine-glutamate cycle	GRIN2B	rs1805476	G=Reduced activity/T=	T/T	imputation DR2=1.00
Glutamine-glutamate cycle	GRM3	rs12704290	A=Increased activity/ G=Reduced activity	G/G	imputation DR2=0.99
Glutamine-glutamate cycle	GRM3	rs6465084	A=Reduced activity/ G=Increased activity	A/A	chip
Glutamine-glutamate cycle	GRM3	rs274622	C=Increased activity/ T=Reduced activity	T/T	chip
Glutamine-glutamate cycle	GRM7	rs3749380	C=Increased activity/ T=Reduced activity	SNP not found in the data	SNP not found in the data (DR2<0.8 imputation)
Glutamine-glutamate cycle	GRM7	rs779867	C=Reduced activity/T=	T/C	imputation DR2=0.95
Glutamine-glutamate cycle	CACNA1C	rs1006737	A=Altered activity/G=Normal activity	G/A	chip
Glutamine-glutamate cycle	GLUL	rs10911021	C=Reduced activity/T=	C/C	chip
Glutamine-glutamate cycle	SLC6A15	rs1545843	A=Reduced activity/ G=Increased activity	A/A	chip
GABA cycle	GABRA2	rs279858	C=Reduced activity/T=	T/T	chip
GABA cycle	GABRA6	rs3219151	C=Increased activity/ T=Reduced activity	C/T	chip
GABA cycle	GABRG2	rs211037	C=Typical activity/ T=Reduced activity	C/C	chip
GABA cycle	GAD1	rs1978340	A=Reduced activity/ G=Increased activity	A/A	chip
GABA cycle	GAD1	rs3749034	A=Increased activity/ G=Reduced activity	G/G	chip
GABA cycle	ABAT	rs2270288	G=Typical activity/ T=Reduced activity	T/T	imputation DR2=0.97
GABA cycle	ALDH5A1	rs2760118	C=Typical activity/ T=Reduced activity	C/C	chip
GABA cycle	GAD2	rs992990	A=Increased activity/ C=Reduced activity	C/A	chip

Via	Gene	SNP	Alleles of the report	Genotype	Source
GABA cycle	GAD2	rs2236418	A=Reduced activity/ G=Increased activity	A/G	chip
GABA cycle	GABRB2	rs1816071	C=Reduced activity/ T=Typical activity	T/T	imputation DR2=0.99
GABA cycle	GABRB2	rs1816072	A=Typical activity/ G=Reduced activity	A/A	imputation DR2=0.99
GABA cycle	GABRB3	rs2081648	C=Increased activity/ T=Reduced activity	T/T	imputation DR2=0.97
GABA cycle	GABRB3	rs4906902	A=Increased activity/ G=Reduced activity	A/A	imputation DR2=1.00
GABA cycle	SLC6A1	rs2697153	A=Reduced activity/ G=Increased activity	SNP not found in the data	SNP not found in the data (DR2<0.8 imputation)
GABA cycle	GABRA1	rs2279020	A=Increased activity/ G=Reduced activity	A/A	chip